Digital Signal Processing Final Project Report

Part I: ECG Signal Denoising for Telemedicine Applications

1. Problem Definition and Specifications

The objective of this part is to design and implement digital filters to remove noise from Electrocardiogram (ECG) signals used in remote health monitoring.

ECG Signal Characteristics

- **Sampling Frequency**: 360 Hz
- **Useful ECG Bandwidth**: 0.5 – 100 Hz

Filtering Objectives

1. **Baseline Wander Removal**: Remove low-frequency drift (< 0.5 Hz) caused by respiration.
2. **Power-line Interference Removal**: Eliminate 50 Hz interference from electrical mains.
3. **Muscle (EMG) Noise Reduction**: Attenuate high-frequency noise (20 – 150 Hz) from muscle activity.

2. Filter Design and Analysis

The following filters were designed to address the specified noise components:

Filter Type	Method	Cutoff Frequency	Objective
High-pass	IIR Butterworth (4th order)	0.5 Hz	Baseline removal
Notch	IIR Notch (Q=30)	50 Hz	Power-line interference
Low-pass	FIR Hamming Window (101 taps)	100 Hz	Muscle noise reduction

Filter Analysis Results

For each designed filter, we analyzed the magnitude/phase response, impulse/step response, and pole-zero diagrams. The results demonstrate:

- **High-pass**: Sharp cutoff at 0.5 Hz with linear phase in the passband.
- **Notch**: Very narrow stopband at 50 Hz to minimize signal distortion.
- **Low-pass**: Smooth transition band and significant attenuation above 100 Hz.

3. Implementation and Validation

The filters were applied sequentially to ECG records 100 and 106 from the MIT-BIH Arrhythmia Database.

Performance Metrics

- **Power Spectral Density (PSD)**: Comparison using the Welch method shows significant reduction in 50 Hz noise and low-frequency drift.
- **SNR Improvement**: Quantitative indicators confirm the effectiveness of the filtering chain.

Record	Calculated SNR (dB)
Record 100	-7.57
Record 106	-3.99

Part II: Multi-Band Speech Equalizer for Podcast Enhancement

1. Objective

To develop a multi-band digital equalizer to enhance speech clarity in podcasts and voice recordings.

2. Program Requirements and Operation

The equalizer operates in **Preset Mode** with the following speech-optimized frequency bands:

- 0–100 Hz (Sub-bass)
- 100–300 Hz (Bass/Warmth)
- 300–800 Hz (Low-mid/Clarity)
- 800–2 kHz (Mid-range/Presence)
- 2–5 kHz (Upper-mid/Sibilance)
- 5–10 kHz (High-mid/Air)
- 10–20 kHz (Highs/Brilliance)

3. Filter Design Options

The system supports both FIR (Window-based) and IIR (Butterworth) structures. For the demo, FIR filters with a Hamming window were used to ensure stable and predictable phase response.

4. Processing and Validation

The processing steps include:

1. Designing band-pass filters for each frequency range.
2. Filtering the input signal through each band.
3. Applying user-defined gains (in dB) to each band.
4. Combining the filtered bands in the time domain.

Results

The spectrogram comparison confirms the boost in the mid-range and high-frequency bands, leading to enhanced speech presence and clarity.

Conclusion

The project successfully implemented digital signal processing techniques for two distinct applications: medical signal denoising and audio enhancement. The results demonstrate the power of digital filtering in improving signal quality and extracting meaningful information from noisy environments.